EE 2010
Fall 2026

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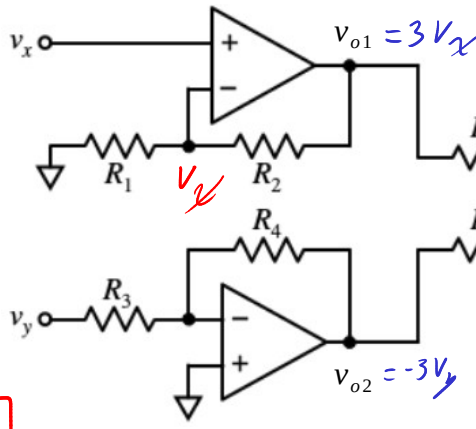
Game 25
10/5/2026

$R_1 = 1\,k\Omega \quad R_2 = 2\ k\Omega \quad R_3 = 5\ k\Omega \quad R_4 = 15\ k\Omega$

$R_5 = 5\ k\Omega \quad R_6 = 1.5\ k\Omega \quad R_7 = 10\ k\Omega$

a) Find v_o in terms of v_x and v_y
Hint: Solve by part. Find v_{o1} in terms of v_x and v_{o2} in terms of v_y first

b) From the v_o equation in part a, find v_o when $v_x = 1\ V$ and $v_y = 0.5\ V$



$V_o = -\left(\frac{R_7}{R_5}(3V_x) - \frac{R_7}{R_6}(3V_y)\right)$

$= -(6V_x - 20V_y)$

$@V_o = 20V_y - 6V_x$

b) $10V - 6V$

$V_o = 4V$

$\frac{V_x - 0}{R_1} + \frac{V_x - V_{o1}}{R_2} = 0$

$\frac{V_{o1}}{R_2} = \frac{V_x}{R_1} + \frac{V_x}{R_2}$

$V_{o1} = \frac{V_x R_2 + V_x R_1}{R_1}$

$= V_x(3)$

$V_{in} = 0$

$\frac{V_{in} - V_y}{R_3} + \frac{V_{in} - V_{o2}}{R_4} = 0$

$\frac{-V_y}{R_3} - \frac{V_{o2}}{R_4} = 0$

$V_{o2} = -V_y \frac{R_4}{R_3}$

$= -3V_y$